BODY CONTROL MODULE (BCM) INSPECTION

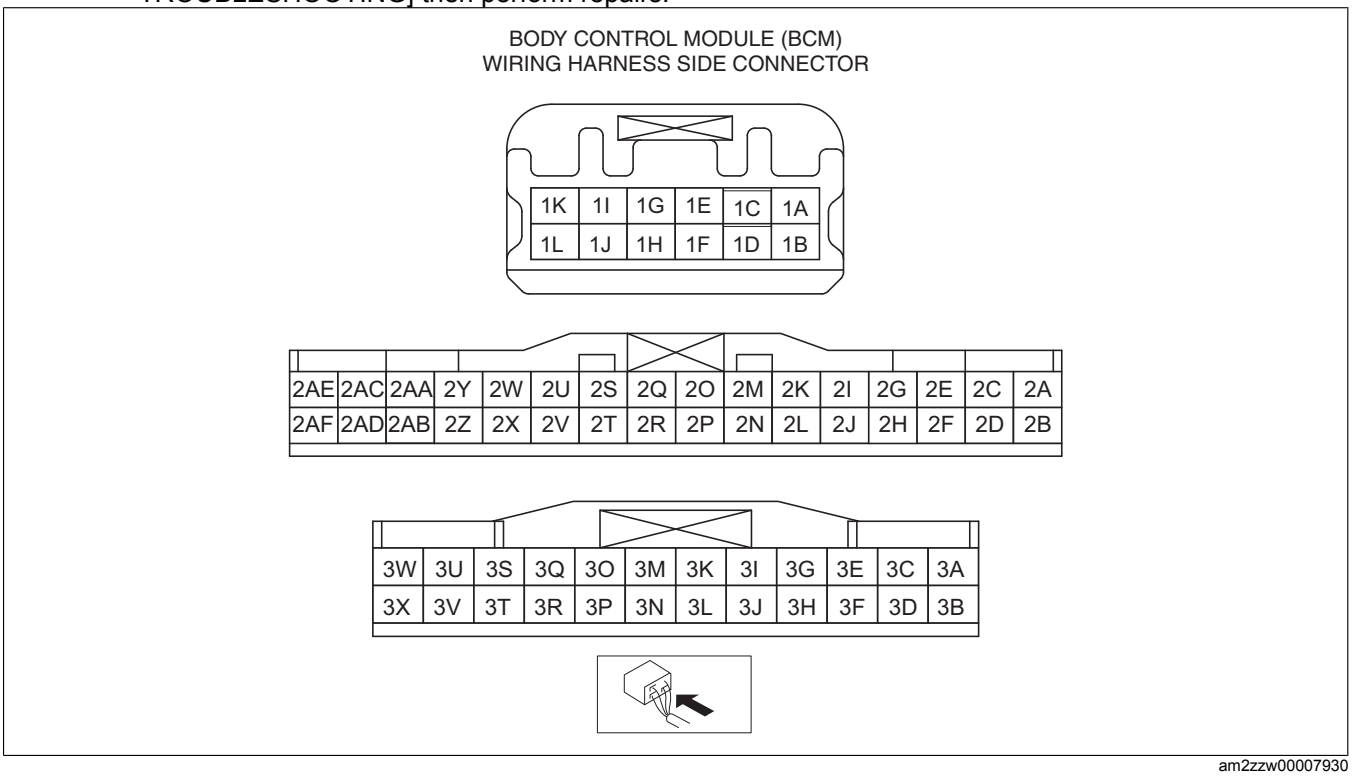
id094000003900

L.H.D.

1. Remove the front scuff plate. (LH) (See FRONT SCUFF PLATE REMOVAL/INSTALLATION.)
2. Remove the front side trim. (LH) (See FRONT SIDE TRIM REMOVAL/INSTALLATION.)
3. Remove the driver-side lower panel. (See DRIVER-SIDE LOWER PANEL REMOVAL/INSTALLATION.)
4. Remove the auto leveling control module (with headlight auto leveling system). (See DRIVER-SIDE LOWER PANEL REMOVAL/INSTALLATION.)
5. Verify that the voltages of each of the terminals are as indicated in the terminal voltage table (reference). (See Terminal Voltage Table (Reference).)
   - If the voltage is not as specified in the terminal voltage table (reference), inspect the parts under Inspection item(s) and related wiring harnesses.
     — If the system does not work normally even though the inspection items are normal, identify the malfunctioning location according to the directions in [DIAGNOSTIC SYSTEM] and [SYMPTOM TROUBLESHOOTING] then perform repairs.

R.H.D.

1. Remove the front scuff plate.(LH) (See FRONT SCUFF PLATE REMOVAL/INSTALLATION.)
2. Remove the front side trim.(LH) (See FRONT SIDE TRIM REMOVAL/INSTALLATION.)
3. Remove the blower unit. (See BLOWER UNIT REMOVAL/INSTALLATION.)
4. Verify that the voltages of each of the terminals are as indicated in the terminal voltage table (reference). (See Terminal Voltage Table (Reference).)
   - If the voltage is not as specified in the terminal voltage table (reference), inspect the parts under Inspection item(s) and related wiring harnesses.
     — If the system does not work normally even though the inspection items are normal, identify the malfunctioning location according to the directions in [DIAGNOSTIC SYSTEM] and [SYMPTOM TROUBLESHOOTING] then perform repairs.



am2zzw00007930

Terminal Voltage Table (Reference)

Terminal	Signal	Connected to	Measurement condition	Voltage (V)	Inspection item(s)
1A	Power supply	HAZARD 15 A fuse	Under any condition	B+	HAZARD 15 A fuse
1B	Power supply	• ROOM 25 A fuse • INTERIOR 15 A fuse	Under any condition	B+	• ROOM 25 A fuse • INTERIOR 15 A fuse
1C	—	—	—	—	—

Terminal	Signal	Connected to	Measurement condition		Voltage (V)	Inspection item(s)
1D*1	Brake light control	Brake light relay	Body control module (BCM) simulation item STOP_LMP_CTL is on		1.0 or less	Brake light relay
			Body control module (BCM) simulation item STOP_LMP_CTL is off		B+	
1E	Rear turn light (RH) control	• Rear turn light (RH) • Side turn light (RH)	Ignition is switched ON (engine off or on)	Turn switch is in right turn position	1.0 or less ↔ B+	• Rear turn light (RH) • Side turn light (RH)
				Turn switch is in position other than above	1.0 or less	
1F	—	—	—		—	—
1G	Rear turn light (LH) control	• Rear turn light (LH) • Side turn light (LH)	Ignition is switched ON (engine off or on)	Turn switch is in left turn position	1.0 or less ↔ B+	• Rear turn light (LH) • Side turn light (LH)
				Turn switch is in position other than above	1.0 or less	
1H	Interior light power supply	Front map light	Under any condition		B+	Front map light
1I	—	—	—		—	—
1J	—	—	—		—	—
1K	Ground	Body ground	Under any condition		1.0 or less	Body ground
1L	Ground	Body ground	Under any condition		1.0 or less	Body ground
2A	Lock input (front door lock-link switch (driver's side))	Front door latch and lock actuator (driver's side)	Driver's door is locked		1.0 or less	Front door latch and lock actuator (driver's side)
			Driver's door is unlocked		Approx. 10	
2B	Unlock input (front door lock-link switch (driver's side))	Front door latch and lock actuator (driver's side)	Driver's door is locked		Approx. 9.5	Front door latch and lock actuator (driver's side)
			Driver's door is unlocked		1.0 or less	
2C	CAN_L	CAN system related module	Terminal used for communication therefore determination based on terminal voltage is not possible.			
2D	CAN_H	CAN system related module	Terminal used for communication therefore determination based on terminal voltage is not possible.			
2E	Rear door latch switch (LH) signal	Rear door latch and lock actuator (LH)	Rear door (LH) is open		1.0 or less	Rear door latch and lock actuator (LH)
			Rear door (LH) is closed		Approx. 9.5	
2F	Rear door latch switch (RH) signal	Rear door latch and lock actuator (RH)	Rear door (RH) is open		1.0 or less	Rear door latch and lock actuator (RH)
			Rear door (RH) is closed		Approx. 9.5	
2G	Front door latch switch (passenger's side) signal	Front door latch and lock actuator (passenger's side)	Front door (passenger's side) is open		1.0 or less	Front door latch and lock actuator (passenger's side)
			Front door (passenger's side) is closed		Approx. 9.0	
2H	Front door latch switch (driver's side) signal	Front door latch and lock actuator (driver's side)	Front door (driver's side) is open		1.0 or less	Front door latch and lock actuator (driver's side)
			Front door (driver's side) is closed		Approx. 9.5	
2I	Serial communication	Power window main switch	Terminal used for communication therefore determination based on terminal voltage is not possible.			
2J	Serial communication	Start stop unit	Terminal used for communication therefore determination based on terminal voltage is not possible.			
2K*2	Door lock switch signal	Door lock switch (RH)	Door lock switch is turned to unlock side (driver's side or passenger's side)		1.0 or less	Door lock switch (RH)
			Door lock switch is turned to lock side (driver's side or passenger's side)		Approx. 1.25	
			Other		Approx 5.0	

Terminal	Signal	Connected to	Measurement condition	Voltage (V)	Inspection item(s)
2L	Front door key cylinder switch signal	Front door latch and lock actuator (driver's side)	Door key cylinder is turned to unlock side	1.0 or less	Front door latch and lock actuator (driver's side)
			Door key cylinder is turned to lock side	Approx. 1.25	
			Other	Approx 5.0	
2M*3	Lock input (front door lock-link switch (passenger's side), rear door lock-link switch)	• Front door latch and lock actuator (passenger's side) • Rear door latch and lock actuator (driver's side), (passenger's side)	Passenger's door or rear seat door is unlocked	1.0 or less	• Front door latch and lock actuator (passenger's side) • Rear door latch and lock actuator (driver's side), (passenger's side)
			Passenger's door and rear seat door are locked	B+	
2N	Rear seat belt switch (LH) signal	Rear buckle switch (LH)	Insert rear seat belt (LH) into buckle	1.0 or less	Rear buckle switch (LH)
			Rear seat belt (LH) not inserted into buckle	B+	
2O	Rear seat belt (center) switch signal	Rear center buckle switch	Insert rear seat belt (center) into buckle	1.0 or less	Rear center buckle switch
			Rear seat belt (center) not inserted into buckle	B+	
2P	Rear seat belt switch (RH) signal	Rear buckle switch (RH)	Insert rear seat belt (RH) into buckle	1.0 or less	Rear buckle switch (RH)
			Rear seat belt (RH) not inserted into buckle	B+	
2Q	Fuel sensor signal	Fuel gauge sender unit	Inspect fuel gauge sender unit and related wiring harness because determination based on terminal voltage is not possible. (See FUEL GAUGE SENDER UNIT INSPECTION.)		
2R	Fuel sensor ground	Fuel gauge sender unit	Under any condition	1.0 or less	Fuel gauge sender unit
2S	• Liftgate unlatch switch signal (5HB) • Trunk unlatch switch signal (5HB)	• Liftgate opener switch (5HB) • Trunk lid opener switch (exterior) (4SD)	• Liftgate is opened (5HB) • Trunk is opened (4SD)	1.0 or less	• Liftgate opener switch (5HB) • Trunk lid opener switch (exterior) (4SD)
			• Liftgate is closed (5HB) • Trunk is closed (4SD)	Approx. 4.5	
2T	—	—	—	—	—
2U	—	—	—	—	—
2V	—	—	—	—	—
2W*3	LIN communication	Theft-deterrent siren	Terminal used for communication therefore determination based on terminal voltage is not possible.		
2X	—	—	—	—	—
2Y	LIN communication	Electrical supply unit (ESU)	Terminal used for communication therefore determination based on terminal voltage is not possible.		
2Z*1	Brake switch signal	• Start stop unit • Brake switch (No.1 signal)	Brake pedal is depressed	B+	• Start stop unit • Brake switch (No.1 signal)
			Brake pedal is not depressed	1.0 or less	
2AA*10	PTC heater communication	PTC heater	Terminal used for communication therefore determination based on terminal voltage is not possible.		
2AB*1	Brake light control	• Brake light relay • Brake light (LH), (RH) • High-mount brake light • Condenser	Brake pedal is depressed	B+	• Brake light relay • Brake light (LH), (RH) • High-mount brake light • Condenser
			Brake pedal is not depressed	1.0 or less	

Terminal	Signal	Connected to	Measurement condition		Voltage (V)	Inspection item(s)
2AC	• Liftgate latch switch signal (5HB) • Trunk lid latch switch signal (4SD)	• Liftgate latch and lock actuator (5HB) • Trunk lid latch and release actuator (4SD)	• Liftgate is opened (5HB) • Trunk is opened (4SD)		1.0 or less	• Liftgate latch and lock actuator (5HB) • Trunk lid latch and release actuator (4SD)
			• Liftgate is closed (5HB) • Trunk is closed (4SD)		Approx. 9.5	
2AD	—	—	—		—	—
2AE	—	—	—		—	—
2AF	—	—	—		—	—
3A	Front turn light (RH) control	Front turn light (RH)	Ignition is switched ON (engine off or on)	Turn switch is in right turn position	1.0 or less ↔ B+	Front turn light (RH)
				Turn switch is in position other than above	1.0 or less	
3B	Front turn light (LH) control	Front turn light (LH)	Ignition is switched ON (engine off or on)	Turn switch is in left turn position	1.0 or less ↔ B+	Front turn light (LH)
				Turn switch is in position other than above	1.0 or less	
3C	CAN_H	CAN system related module	Terminal used for communication therefore determination based on terminal voltage is not possible.			
3D	CAN_L	CAN system related module	Terminal used for communication therefore determination based on terminal voltage is not possible.			
3E	Blower motor relay control	• Blower relay • HEATER 40 A fuse	Ignition is switched ON (engine off or on)	Blower motor is operated using fan switch	1.0 or less	• Blower relay • HEATER 40 A fuse
				Blower motor is turned off by pressing climate control unit OFF button	B+	
3F	Windshield wiper INT relay control	Wiper INT relay	Ignition is switched ON (engine off or on)	Windshield wiper switch is in position other than OFF	1.0 or less	Wiper INT relay
				Windshield wiper switch is in OFF position	B+	
3G	Windshield wiper motor (HI) control	Wiper HI relay	Ignition is switched ON (engine off or on)	Windshield wiper switch is in HIGH position	1.0 or less	Wiper HI relay
				Windshield wiper switch is in OFF position	B+	
3H	Horn relay control	Horn relay	Ignition is switched ON (engine off or on)	Horn switch is on	1.0 or less	Horn relay
				Horn switch is off	B+	
3I*3	Front fog light relay control	Front fog light relay	Ignition is switched ON (engine off or on)	Front fog light switch is on	1.0 or less	Front fog light relay
				Front fog light switch is off	B+	

Terminal	Signal	Connected to	Measurement condition		Voltage (V)	Inspection item(s)
3J	Head light cleaner relay control	• Head light cleaner relay • H/CLEAN 20 A fuse	Ignition is switched ON (engine off or on)	Connect the M-MDS and turn on the "H/L_CLN_RY" using the body control module (BCM) simulation function	1.0 or less	• Head light cleaner relay • H/CLEAN 20 A fuse
				Connect the M-MDS and turn off the "H/L_CLN_RY" using the body control module (BCM) simulation function	B+	
3K	Rear window defogger relay control	Rear window defogger relay	Ignition is switched ON (engine off or on)	Rear window defogger switch is on	1.0 or less	Rear window defogger relay
				Rear window defogger switch is off	B+	
3L*4	Burglar horn relay control	Theft-deterrent relay	Ignition is switched ON (engine off or on)	Horn switch is on	B+	Theft-deterrent relay
				Horn switch is off	1.0 or less	
3M	Autostop switch signal	Windshield wiper motor	Ignition is switched ON (engine off or on)	Windshield wipers are stopped in position other than park	1.0 or less	Windshield wiper motor
				Windshield wipers are stopped in park position	B+	
3N	Power supply	• ENG+B 7.5 A fuse • ROOM 25 A fuse	Under any condition		B+	• ENG+B 7.5 A fuse • ROOM 25 A fuse
3O	Ignition power supply	METER1 10 A fuse	Ignition is switched ON (engine off or on)		B+	METER1 10 A fuse
			The ignition is switched off (LOCK)		1.0 or less	
3P	—	—	—		—	—
3Q*5	Washer fluid level sensor signal	Washer fluid level sensor	Ignition is switched ON (engine off or on)	Washer fluid level sensor is less than MIN (washer fluid level sensor on)	1.0 or less	Washer fluid level sensor
				Washer fluid level sensor is MIN or more (Washer fluid level sensor off)	B+	

Terminal	Signal	Connected to	Measurement condition		Voltage (V)	Inspection item(s)
3R	Brake fluid level sensor signal	Brake fluid level sensor	Ignition is switched ON (engine off or on)	Brake fluid level sensor is less than MIN (brake fluid level sensor on)	1.0 or less	Brake fluid level sensor
				Brake fluid level sensor is MIN or more (brake fluid level sensor off)	B+	
3S	LIN communication	Current sensor	Terminal used for communication therefore determination based on terminal voltage is not possible.			
3T	Horn switch control	Horn switch	Ignition is switched ON (engine off or on)	Horn switch is on (long press)	1.0 or less	Horn switch
				Horn switch is off	B+	
3U	Trunk unlatch switch signal (interior)	Trunk lid opener switch (interior)	Trunk lid opener switch (interior) is on (trunk lid opener switch (interior) is pressed)		1.0 or less	Trunk lid opener switch (interior)
			Trunk lid opener switch (interior) is off (trunk lid opener switch (interior) is not pressed)		B+	
3V*7	LIN communication	DC-DC converter	Terminal used for communication therefore determination based on terminal voltage is not possible.			
3W	Hood latch switch signal	Hood latch switch	Hood is open		Approx. 9.0	Hood latch switch
			Hood is closed		1.0 or less	
3X	LIN communication	• Auto-light sensor/rain sensor*8 • Theft-deterrenet sensor*9	Terminal used for communication therefore determination based on terminal voltage is not possible.			

*1 : Vehicles with smart city brake support (SCBS)
*2 : Vehicles with door lock switch
*3 : Vehicles with door lock link switch
*4 : Vehicles with theft-deterrent horn
*5 : Vehicles with washer fluid level sensor
*6 : Vehicles with trunk lid opener switch (interior)
*7 : Vehicles with i-stop system/without i-ELOOP
*8 : Vehicles with auto light system/auto wiper system
*9 : Vehicles with theft-deterrent siren
*10 : Vehicles with PTC heater